STRANGE NEW WORLDS: THE SEARCH FOR M-CLASS PLANETS

ALICE LUNDGREN, EMMALINE MCKINNON, FLETCHER MURRAY, ASHLEY SPENCER, JARED WINN

Abstract. We work to detect and locate inhabitable, M-class planets outside our solar system. In order to identify and describe exoplanets, we clean and analyze the time series data of a star's electromagnetic emissions. Methods include binary classification strategies, time series decompositions, ARMA, and Kalman Filters. While some methods are unsuccessful, the time series decomposition produces a qualitative way to predict planetary existence. Additionally, the Welch algorithm determines that the planets of interest are outside the habitable zone. The Kalman filter predicts transit spike timing, but was unable to predict the correct orbital position of the planet.

1. Research question and overview of the data

The universe is vast and the majority of it is yet to be explored. Except for the small region we call the solar system, we cannot explore it with physical probes; instead, we must content ourselves with gathering observations from on or nearby Earth. Given these difficulties, one of the principal methods used by astronomers to identify planets is studying electro-magnetic emissions of the stars they orbit [Natnd]. A planet itself will not emit in the electromagnetic spectrum; however, when it passes between the star and the observer, it will decrease the intensity of the star's perceived emissions for a period of time. Using these measurements, coupled with information about the type and distance of the star, scientists can not only identify possible exoplanets, but also predict attributes such as their size and orbit, information critical to answering the question of whether or not they may be able to sustain carbon-based lifeforms.

The Kepler Space Telescope was launched in 2009 with the mission of hunting for exoplanets by measuring the stellar brightness of more than 150,000 stars [Nat] over an extended period of time. The collected data was beamed back to earth and extensively analyzed in conjunction with other characteristics of the stars. In our project, we analyze the Kepler brightness data, known as light curves, and attempt to reproduce some of the results found by physicists and astronomers.

We have three main goals:

Date: 16 April 2026.

(1) Identify if an exoplanet exists.
(2) Determine if a planet lies in a habitable zone.
(3) Pin the orbital location of an exoplanet at a given time.

As members of Starfleet's scientific division, our mission is to detect M-class planets and precisely determine their locations for future settlement efforts.

2. Data Cleaning / Feature Engineering

NASA stores and makes the Kepler mission data freely available [NAS17]; we use light curves and a Kaggle dataset [kep] which reports information about the size, orbit, and other characteristics of possible exoplanets. Each possible exoplanet has a unique Kepler ID, allowing us to query the NASA database using the package `lightkurve` to access its observed light curve. For each of the Kepler IDs we obtain the time series flux measurements as well as the sensor's uncertainty measurements.

Because the Kepler telescope was solar powered, it performs a roll every quarter in order to keep its panels aligned. As a result, target stars are observed by a different photo-sensor every quarter. Measurements of the same object are statistically different enough that each quarter is treated as a stand alone time series and cannot be reliably combined with observations of the same target taken during a different quarter[NAS17].

2.0.1. *De-noising.* The Kepler data is very noisy due both to measurement failures and the inherent variability of star brightness. We experiment with several methods in order to denoise the data:

- Rolling Windows: The rolling window filter is simply a moving average; we experiment with window lengths between two and nine.
- Savgol Filter: The Savitzky-Golay filter works by approximating a window of the data with a low degree polynomial using least squares. This creates a smooth and continuous signal. We experiment with polynomials of order 2 to 4 and windows of size 2 to 9.
- Median Filter: The median filter works by sliding a window and replacing the center value with the median of everything contained within the window.
- Wavelets: Wavelet filters decompose signals into pulses and drop those whose frequencies fall beyond a threshold. For our purposes, we use an empirically derived threshold coefficient of 20.97. Figure 1 shows a lack of pattern in the residuals, indicating successful denoising.

Wavelets produce the best results, so we use it as our chosen denoising method for binary classification.

3. Data Visualization and Basic Analysis

The data returned by the Kepler telescope takes the form of a series of pixel patches, with the target star roughly centered in the middle of the

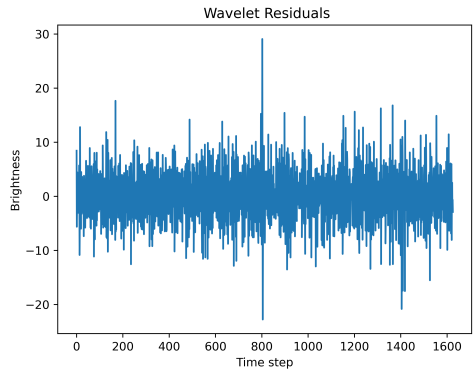
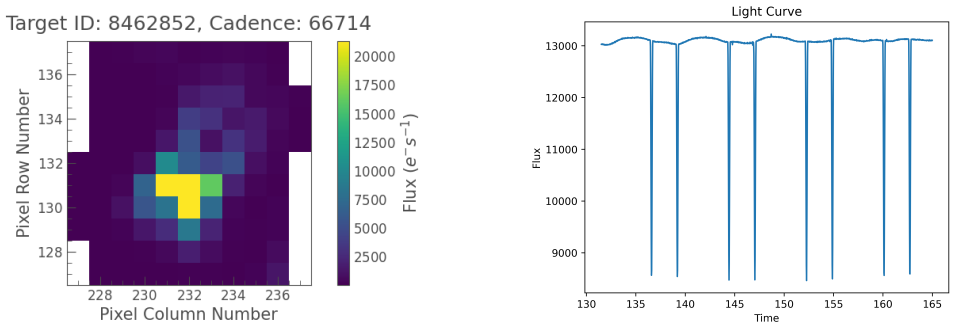


Figure 1. Residuals from wavelet decomposition. The lack of pattern indicates successful denoising.

patch as seen in Figure 2a. In order to create a light curve we collapse each patch by summing the pixel values to create a one dimensional signal.



(a) Pixel file (b) Flux

Figure 2. The raw pixel image of a star's brightness at a given time and its flux over 35 days

The main method of using light curves to determine the existence of an exoplanet is known as the Transit Method[Unia]. A periodic dip in observed flux indicates a possible exoplanet passing between the observer and star.

Surprisingly, clean measurements with a clear periodic nature seen in Figure 2b, were often false positives; this points to more complex physical laws than we first supposed.

4. Learning Algorithms and In-depth Analysis

4.1. **Mission One: Detecting an Exoplanet.**

4.1.1. *Binary Classifiers.* To address our first question of the existence of an exoplanet, we first use a random forest classifier.

Table 1. Random Forest Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	565	0
Actual Negative	5	0

The huge imbalance in the data [kep] leads to the model predicting that there are no planets, as seen in Table 1. Increasing both the number of estimators and their depth did not change the result. As such, we try more subtle methods. We begin with an SVM using `sklearn.svm.SVC`. Scaling the data using a standard scaler does not improve results. We then try XG Boost's classifier, tweaking the importance scaling parameter to reflect the expected number of true positives. Once again, this does not improve performance. Variations of the evaluation metric and tree parameters yield similar results. As seen in Figure 3, two-component PCA does not reveal any structure or separation between the stars with or without planets.

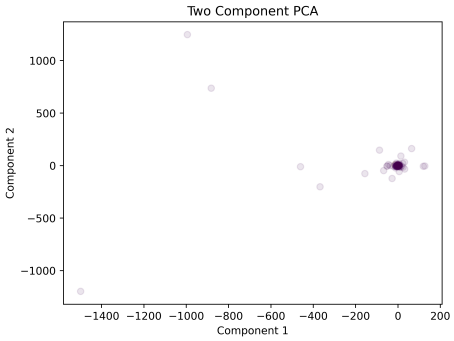


Figure 3. PCA dimensionality reduction.

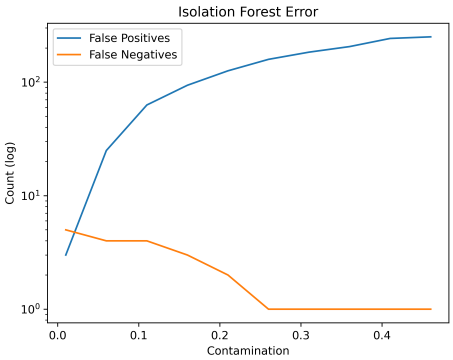
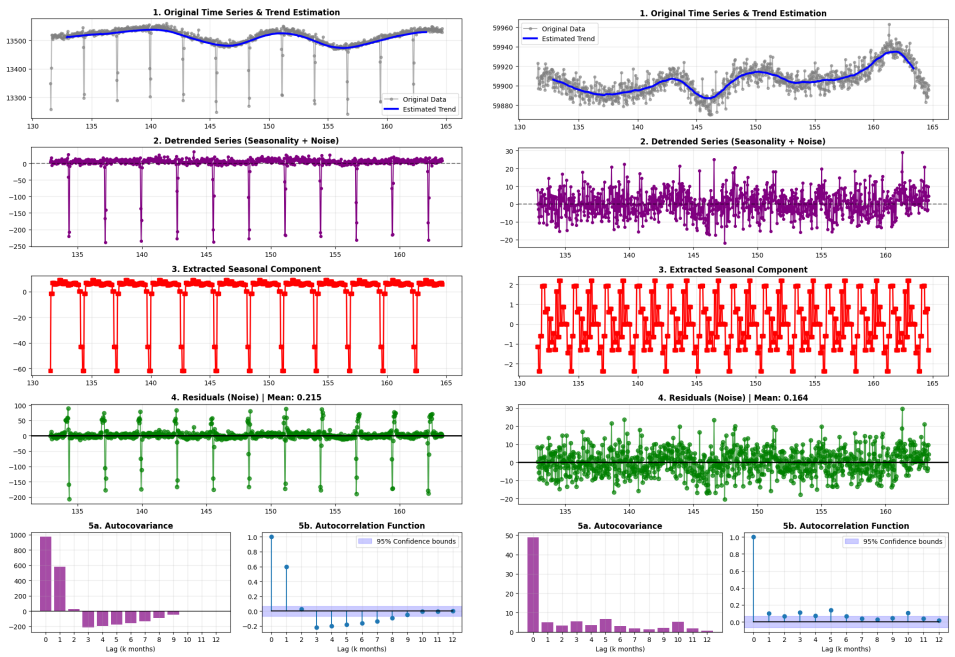


Figure 4. Isolation Forest error evolution.

Instead of training on both classes, we switch to an anomaly detection approach, where we train on the dominant class (no planets) and then try to detect any deviations from the normal that might indicate the presence of a planet. We work with isolation forests, single-class SVMs, and local outlier factor; however, as we tweak the parameters, it became clear that the signals corresponding to planets are not distinct from those without. Figure 4 shows that to correctly identify any exoplanets, we falsely classify a minimum of 43 non-planets.

4.1.2. *Time Series Decomposition.* We utilize a time series decomposition to qualitatively inspect the existence of an exoplanet. As the planet passes in front of the star, the light dims. This will happen once every orbital period. Thus, in the light flux time series decomposition, if there is a planet, there

will be a strong seasonal component. This is showcased in Figure 5a. Stars that do not have an exoplanet, theoretically, will not have a strong pattern in the seasonal component as seen in Figure 5b.



(A) Star with an exoplanet (B) Star without an exoplanet

Figure 5. Comparison of time series decomposition of stellar light flux for systems with and without exoplanets.

Though this is not an airtight method, a simple time series decomposition is a good place to start to predict the existence of an exoplanet.

4.2. **Mission Two: Determining Habitability.** Every star has a zone of habitability. In the habitable zone, the temperature is such that water can be found in liquid form, a necessary precursor to the maintenance of life. Because the habitable zone is determined by temperature, it can be calculated from the radius and effective temperature of a star. Additionally, Kepler's third law of planetary motion states that the orbital distance of a planet from its star is proportional to its period:

$$r = \sqrt[3]{\frac{GMT^2}{4\pi^2}} \tag{1}$$

where T is a planet's period, M is the mass of the star, G is the universal gravitational constant, and r is the orbital radius. Hence, the period of a planet's orbit can be used to calculate its orbital radius and determine if it lies in the habitable zone. Due to time and space constraints, we restrict

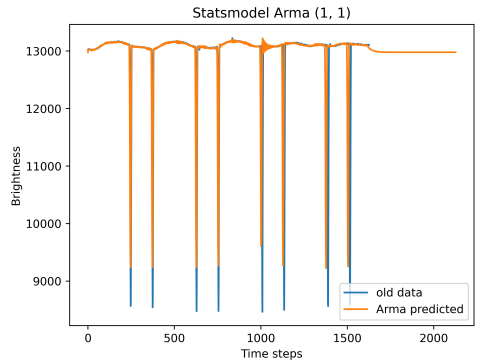


Figure 6. Results of best performing ARMA model

our search to five sample light curves and examine two ways of determining a planet's period from this data.

4.2.1. *ARMA.* If we can predict a pattern in the flux, then the dips are not a product of noise, but instead an indicator of the period of our orbit. We run a parameter sweep to identify the optimal p and q for our $ARMA(p, q)$ model using the model's Akaike Information Criterion score. Unfortunately, even the highest-scoring model does not accurately predict future observations. as seen in Figure 6.

4.2.2. *Welch's Algorithm.* We use `scipy.signal.welch()` on time series data of a star's flux to find its orbiting planet's period. `scipy.signal.welch()` uses the discrete Fourier transform to transform a signal into its constituent frequencies and the respective magnitude of their powers. Assuming the frequency at which the star's measured flux is obscured by an exoplanet is one of the dominant frequencies in a signal, we can extract the frequency in the signal with the most power. Using the fact that $T = \frac{1}{f}$, we obtain the period of the orbiting planet. From Kepler's Third Law of Planetary Motion (Eq 1), we calculate the orbital radius. Results are in Table 2.

Our estimations reveal that all the exoplanets examined fall well outside the habitable zone.

Table 2. Welch Algorithm Predictions (†No Exoplanet)

Kepler ID	Period (days)	Orbital Radius (Solar Radii)	Habitable Zone (Solar Radii)
11446443	0.17	1.3	270 - 390
11414511	5.23	12.5	1562-2251
9579641	5.23	12.5	2607-3756
10583181†	1.3	4.9	2077-2993
8299955†	5.23	12.5	1888-2720

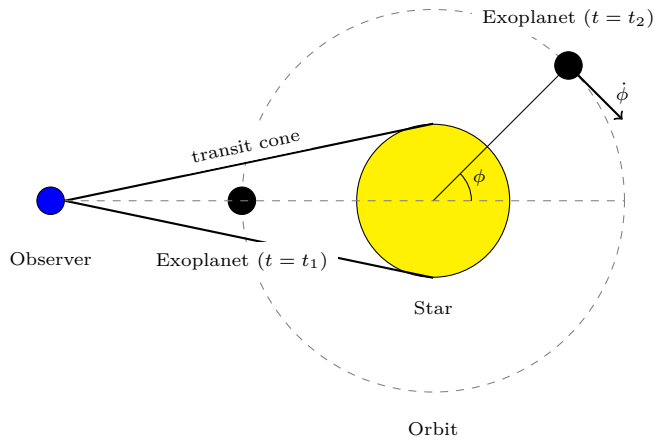


Figure 7. State-space model of exoplanetary system. The transit cone is shown relative to an observer at $t = t_1$. At $t = t_2$, we see the exoplanet is defined by its phase ϕ and orbital velocity $\dot{\phi}$

4.3. **Mission Three: Predict Location of Exoplanet.**

4.3.1. *State Space Model.* In this section, we develop a simple state-space model of the system. We then leverage the flux observations and Kalman Filter to estimate the location of a possible planet in its orbit.

The state space shown in Figure 7 assumes a co-planar observer and an exoplanet with a circular orbit. The exoplanet's state is described by $x = [\phi, \dot{\phi}]$, representing its phase (angle from stars center) and orbital velocity respectively. We update the state of the exoplanet as a noisy physics model with constant velocity where w_{1_k} and w_{2_k} represent the normally distributed noise.

$$\phi_{k+1} = \phi_k + \Delta t \dot{\phi}_k + w_{1_k}$$
$$\dot{\phi}_{k+1} = \dot{\phi}_k + w_{2_k}$$

The view of an observer is defined by a *transit cone* (see Figure 7). When an exoplanet enters this cone, it partially obscures the star and results in a dip in brightness. We will model this dip in brightness as a periodic function of the phase $H : \phi \to \mathbb{R}$. The function must approximate a light curve that transitions from a steady state to a smooth minimum and back [Unia], ensuring it accurately models the transit depth and steady-state magnitudes for a reasonable range of ϕ [Unib]. A good fit for such a function is a gaussian dip subtracted from a baseline:

$$H(\phi) = B - d \exp\left(\frac{-1 + \cos(\phi - \phi_0)}{s}\right) \tag{2}$$

where B represents average flux without a transit, d is the depth of drop in flux during a transit, s is the duration or width of the transit, and ϕ_0 is the phase of maximal obstruction.

Note that we include $\cos(\phi - \phi_0)$ in the dip to ensure periodicity of our transits, and that the parameters of this function must be estimated from the data for each exoplanetary system.

Thus, we can write our full model as

$$x_k = Fx_{k-1} + w_k$$
$$z_k = H(\phi) + r_k$$

where z_k are the observed flux measurements, $w_k \sim N(0, Q)$, $r_k \sim N(0, \sigma_r)$ and

$$F = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix} \qquad Q = \begin{bmatrix} \sigma_{w_1}^2 & 0 \\ 0 & \sigma_{w_2}^2 \end{bmatrix}$$

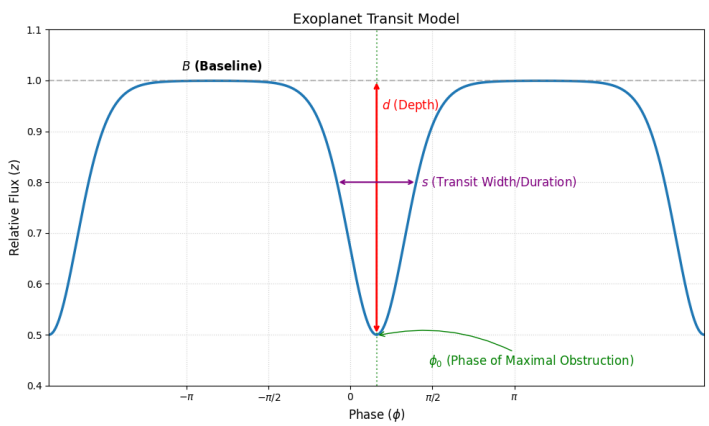


Figure 8. Graph of observation function $H(\phi)$. We see as a planets phase ϕ increases it periodically leads to dips in brightness

Due to the nonlinearity of our observation function H we cannot apply the standard Kalman filter. In the following sections, we attempt two methods for handling this nonlinearity: the extended Kalman filter and the unscented Kalman filter.

4.3.2. *Extended Kalman Filter.* We use an extended Kalman filter (EKF) to account for the nonlinearity of the observed flux measurements, as defined in Equation 2. The EKF linearizes the model by calculating the first-order Taylor expansion using the Jacobian of the non-linear function.

Flux measurements are standardized by subtracting the mean and dividing by the standard deviation before utilizing an EKF as seen in Figure 9. For every iteration of EKF, the phase, or angle of the planet from the star's

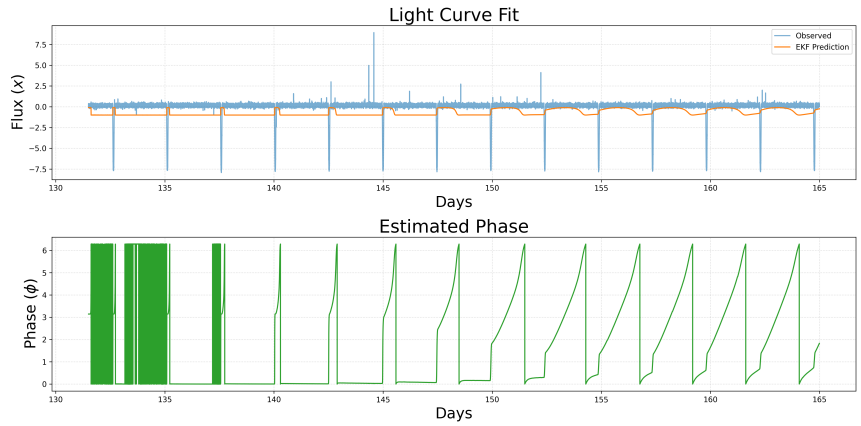


Figure 9. The extended Kalman filter tracks the phase and flux of Planet 11446443 with $s = 0.9$ and $d = 1.0$ from Equation 2.

center, is wrapped modulo 2π to ensure it remains in the interval $[0, 2\pi)$. After approximately 15 time intervals, the phase measurements snap into a rhythm that oscillates through the whole range of $[0, 2\pi)$. The extended Kalman filter correctly maps the timing of each flux spike but does not achieve the correct depth.

Since our state space model does not capture the seasonal trend in flux, we attempt to run EKF on the detrended data of stars with known exoplanets, as found in Figure 5. The results do not predict the time of the flux spikes or their depth.

4.3.3. *Unscented Kalman Filter.* The extended Kalman filter relies on local linearization to ensure propagation of the gaussian distribution. However the steep derivatives in our observation function leads to numerical instability. In constrast, the unscented Kalman filter (UKF) represents our gaussian prior using a finite set of sample points. These are then sent through the nonlinear functions to capture the distribution of the posterior [WVDM].

We utilize `filterpy.kalman.UnscentedKalmanFilter()` with Merwin scaled sigma points as our samples [WVDM]. As before, we normalize the flux measurements and run the filter on the detrended data. Despite some initial burn-in, the filter manages to replicate the timing of large dips in flux, albeit without the same depth. However, investigation of the estimated phase yields nonphysical results. Despite starting in a smooth orbit, the state estimate eventually dwells at the ingress point and rapidly accelerates in and out of the transit window to replicate the sudden dips in flux. A video of this evolution can be found here.

Both EKF and UKF fail to give the expected results for estimating the current phase of the planet. The advanced nature of our methods means

they can't possibly be wrong; thus, we ask Starfleet to send a scientific mission to investigate the orbital anomalies in these systems.

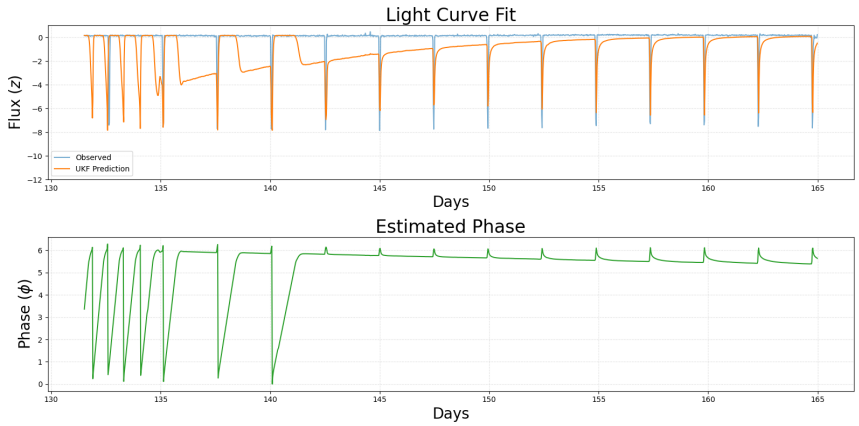


Figure 10. The unscented Kalman filter ($B = 0.16$, $\phi_0 = 0$, $s = 0.08$, $d = 8$) accurately tracks transit timing but exhibits nonphysical phase "snapping" to replicate flux dips.

5. Ethical Implications and Conclusions

We source our data from Kaggle and the public-facing NASA data repository, so we do not need to worry about PII or licensing issues. While our research does not yet yield enticing results, a future space-faring galactic empire might misuse our insights to conquer and oppress less advanced inhabitants of other worlds.

As such we solemnly swear that we will not use the results of this research for a period of no less than five standard earth years to establish any form of evil galactic empire or otherwise enslave the galaxy. We politely ask that any who choose to commit such atrocious acts cite our paper.

Our findings are incomplete but promising. Regarding mission one, identifying exoplanets, binary classification algorithms yield little success. However, the seasonal component of a time series decomposition can indicate an exoplanet. Regarding mission two, assessing habitability, the ARMA model fails to predict interesting results. Welch's algorithm predicts that our identified planets are not in the habitable zone. Pushing forward, we use the Kalman filter to pin the orbital location of the planets. This strategy yields a prediction of strange teleportation behavior. As Starfleet researchers, we are convinced that habitable worlds await discovery beyond our current charts. We will continue to analyze stellar flux signatures, refining our models with each iteration, until we identify those rare and life-sustaining M-class planets among the stars.

References

[kep] keplersmachines. Kepler labelled time series data.

[NAS17] NASA. Kepler exoplanet search results. `https://www.kaggle.com/datasets/nasa/kepler-exoplanet-search-results`, 2017. Accessed: 2026-03-16.

[Nat] National Aeronautics and Space Administration. Kepler.

[Natnd] National Aeronautics and Space Administration. The transit method. NASA Science: Nancy Grace Roman Space Telescope, n.d. Accessed: April 16, 2026.

[Unia] University of Nebraska-Lincoln Astronomy Education. Transit depths and light curves.

[Unib] University of Nebraska-Lincoln Astronomy Education. Transit duration and inclination.

[WVDM] Eric A. Wan and Rudolph Van Der Merwe. The unscented kalman filter for nonlinear estimation.

6. Code

All of our code can be found at:

https://github.com/Jaredwinn77/ACME_VOL3_PLANET_FINDER